This work acknowledges Prof Chris van Schoor's contribution in compiling this summary 2017

Data group 1: Specify PRODUCTS and NUMBERS/Volumes to be manufactured or assembled

- Specify the product design, material input and packaging requirements, etc.
- Specify QUALITY requirements and expected SCRAP rates
- Calculate the required DAILY OUTPUT and capacity per product and component to meet customer demand (how much, how fast, when, in what form)
- Material cost estimates

(Customer demand, Daily production plans, Component and assembly drawings, parts list, BOMs (Bill of Materials), product structure diagram, assembly chart)

Data group 2: Specify the Primary Manufacturing (including Material handling) PROCESSES ACTIVITIES as well as Support Activities

- Determine required PROCESSES and operations activities
- Determine ORDER and LOT sizes (what is a unit?)
- Distinguish between Fabrication and Assembly ACTIVITIES
- Identification and evaluation of EQUIPMENT
- Establish personnel requirements for operating the equipment
- Estimated time standards for set-up and operation, number of machines required, cycle times, equipment and process efficiencies and utilisation, and requirements for assembly line balancing
- Estimated capital equipment costs
- Determine manufacturing SERVICE requirements (electricity, water, fuel, gas, waste disposal, etc.) for each process

(Operation charts or procedures, process flow charts, route sheets, schedules, material handling requirements, Activity charts, Activity/Precedence diagrams)

- Considering MATERIAL HANDLING and available EQUIPMENT based on weights, specifications and dimensions of raw materials, products and finished goods. Employ the questioning approach to develop alternative concept material handling system designs.
- Specify the equipment, unit loads, capacity, etc. of the recommended system.

(Questioning approach answers, Material handling planning chart, Equipment specifications)

- Determine requirements for RAW MATERIAL, WIP and FINAL PRODUCT stores/STORAGE areas based on the specified inventory control and production planning and control approach or policies.
- Determine requirements for RECEIVING and DISPATCH areas. Visualize how materials will flow into and through and out of the facility.
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- Determine requirements for OFFICES and PERSONNEL services, based on organisation structure, direct and indirect labour and admin personnel. Also think of PARKING spaces and how the flow of people into and out of the facility work.
- Establish the scope of other SUPPORT activities such as security, maintenance, tool storage and control, cleaning, etc.

(Production planning, control and flow specifications, order and lot sizes, storage volumes and storage min/max policies, specification of requirements for offices and support activities)

Data group 3: Determine the RELATIONSHIPS between Activities

- Quantify and visualise (map) flow relationships if possible
- Determine Activity relationships (non-quantifiable)

(From-to-charts, Activity relationship chart and diagram, Space relationship diagram, String and/or spaghetti diagrams)

Data group 4 : Determine the SPACE Requirements for all Activities

- Establish the SPACE requirements for individual work centres/departments/functions, considering requirements for equipment, materials and personnel including assembly lines etc as determined by the processes as well as all support activities and services.
- A LAYOUT SKETCH (BLOCK LAYOUTS) of each work centre is required.
- Combine work stations/centres in departments and design AISLES between work stations/centres to optimise material and personnel movement and space utilisation.

(Work station/block layouts including aisles, departmental area and service requirements sheets)

- Determine and motivate the size of storage areas for input materials, work-in process and finished products, considering the type of storage equipment, store location methods, etc. The layout of dedicated storage areas and storage equipment must be specified.

(Storage Analysis Chart, Storage equipment specifications, STORES layout)

- Determine the requirements for receiving and shipping areas and specify the layout and facilities, including docks, vehicle loading and holding areas, vehicle manoeuvring and parking.

(Receiving and Shipping Analysis Chart, RECEIVING and SHIPPING area layout)

- Determine the space required for offices, rest rooms, parking and other personnel services as required by the operations for these areas.
- Functionally design each area with simple layouts.

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- Determine total space requirements for different areas (sum up requirements).

Data group 5: Develop ALTERNATIVES Facilities Layout Plans, EVALUATE and SELECT to get to Master Facility Plan (MFP)

- Integrate results of previous phases in a summarised format as basis for assumptions
- Develop basic layout configurations for a number of alternatives (at least 3), with consideration of practical limitations and adjustments
- Basis for alternatives can include things such as: product versus process orientation, different flow patterns, alternative materials handling systems, etc. Also remember that alternatives do not need to be the whole facility, it can also just include certain work areas, as it makes common sense.

(Block layouts alternatives,)

- Evaluate alternatives against predetermined and weighed criteria.

(Evaluation matrix, Weighted criteria or Pro/Con analysis, Structured motivation for product and process design, space requirements and relationships, equipment choices)

- Select and motivate the proposed/preferred alternative
    - Master plan (*detailed final layouts of site and buildings*)
    - Estimated cost of facility and total manufacturing cost

(Master plan of site and building layout, Summary of area requirements and specified equipment)